

Relational-Indexed RAG Architecture

OVERALL SYSTEM DESIGN & SCHEMA SPECIFICATION

VECTOR DATABASE SAMPLE

The vector database serves strictly as the entry point for semantic queries, mapping high-dimensional embeddings directly to relational records via foreign keys without storing bulky text payloads.

```
{
  "id": "policy-1042",
  "title": "Remote Work Policy",
  "content": "Employees can work remotely...",
  "department": "HR",
  "country": "India",
  "foreign_key": "foreign_key",
  "embedding": "title of the document in the database vector representation"
}
```

MYSQL TABLE SCHEMA

The authoritative source of truth where full content and detailed metadata are stored and managed relationally.

```
CREATE TABLE policies (
  id VARCHAR PRIMARY KEY,
  title VARCHAR,
  content TEXT,
  department VARCHAR,
  country VARCHAR
);
```

SYSTEM ARCHITECTURE & WORKFLOW

- **Semantic Entry Point:** The vector database handles initial query ingestion and embedding similarity scanning.
- **Relational Hydration:** Upon scanning embedding potential, the system follows the foreign key relationship to detect and disclose the right content from the relational database.
- **Entry-Only Storage:** The vector database is used strictly for entry queries and index lookups, not for output storage.
- **Ingestion & Sync (Sniffer):** New records and updates are tracked and synchronized via a custom database sniffer pipeline.
- **Hydration & Ranking Engine:** Python execution logic coordinates directly with the relational database through dynamic SQL execution scripts to fetch payloads.
- **Access Control & Filtering:** Security metadata fields are appended during vector database creation to enforce precise filtering at retrieval time.

Author: Laurie MAVOUNGOU
Phone: +242055417936
Email: lmavoungou@outlook.be

Upcoming CEO & Founder @ JK AI | AI Security Researcher
Former Systems Engineer | Fulbright Alumna

Special Thanks & Bibliography: Gratefully inspired by concepts explored in "[Vector Databases Demystified](#)" by **Abhinav Verma** (Published on [Medium](#)).